

Arc-and-Tumble Dynamics

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1 Introduction

Microorganisms use **run-and-tumble** motion as a strategy to navigate through environments. This allows the organism to collect information along its trajectory, while tumbling allows it to change direction towards another region.

Arc-and-tumble motion is a form of *active* particle motion in which the particle undergoes *self-propelled* motion along a *curved* trajectory. Between tumbles, the particle maintains its circular motion. At a tumble, its orientation changes randomly, after which it continues along a new circular arc.

In this project, we develop the dynamics of an arc-and-tumble trajectory with **velocity** v along a circular arc of a constant **radius** ρ , while tumbles occur at a constant **tumble rate** α . The probability of a tumble during a **time interval** Δt is defined by the Poisson relation:

$$P_{\text{tumble}} = 1 - e^{-\alpha \Delta t},$$

Thus, between tumbles, the position and orientation of the particle evolve *deterministically* along an arc. At a tumble, the orientation θ changes *randomly*, introducing the stochastic component of the motion.

2 Equations of Motion

The position of the particle r is described by the two-dimensional Cartesian vector

$$\mathbf{r}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}.$$

The direction of motion of the particle is described by the orientation angle $\theta(t)$. The corresponding unit orientation vector is

$$\hat{\mathbf{e}}(t) = \begin{pmatrix} \cos \theta(t) \\ \sin \theta(t) \end{pmatrix}.$$

For a circular arc of radius ρ , the arc length is

$$s = \rho \theta,$$

and the velocity along the arc is

$$v = \frac{ds}{dt} = \rho \frac{d\theta}{dt},$$

so that

$$\omega = \frac{v}{\rho}.$$

Integrating the angular velocity with respect to time gives the orientation angle

$$\theta(t) = \theta(0) + \int_0^t \frac{v(\mathbf{r}(\tau))}{\rho} d\tau.$$

where τ is the dummy time variable and

$$\mathbf{r}(t) = \int_0^t v \hat{\mathbf{e}}(\tau) d\tau.$$

2.1 Mean Squared Displacement (MSD)

To quantify how a particle or an ensemble of particles spreads with time, we calculate the **mean squared displacement (MSD)**. The MSD provides a measure of how an ensemble of particles or trajectories spreads with time and thus determines how effectively they explore the surrounding space.

The MSD can be obtained from the squared displacement of a single particle p at time t_n , defined as

$$r_p^2(t_n) = (x_p^n)^2 + (y_p^n)^2.$$

For N particles,

$$\text{MSD}(t_n) = \frac{1}{N} \sum_{p=1}^N r_p^2(t_n),$$

where N is the total number of particles or trajectories.

3 Objective

The objective of this project is to develop a Python code for this arc-and-tumble motion and apply the programming and numerical tools introduced in the Scientific Python course.

The simulation will be used to first generate an individual particle trajectory and then expand it to an ensemble to calculate the mean squared displacement (MSD) for the constant-velocity case.

The project should demonstrate your ability to:

1. identify and define the physical and numerical parameters required by the model by extracting data from published work or making informed guesses;
2. use appropriate Python data types and data structures to organise the simulation;

3. use conditional statements and loops;
4. define and use functions to organise repeated calculations;
5. use NumPy and SciPy for numerical calculations and analysis;
6. use Pandas to organise, store, and analyse simulation data;
7. use plotting libraries to visualise the data.

4 Project Tasks

Each group is required to complete Tasks 4.1–4.4. Choose between Tasks 4.5 and 4.6. If two groups do this project, they CANNOT submit the same task.

4.1 Model Parameters

The physical parameters for the arc-and-tumble model will be obtained from (Turuban, Noselli, Beran, & DeSimone, 2025). Read the paper and determine appropriate values for the particle velocity v , the radius ρ of the circular trajectory, and the tumble rate α . If you cannot find the relevant data, justify your selection of parameters.

4.2 Single Particle

Develop a code using the equations above for the motion of a single arc-and-tumble particle.

At each time step, your program should calculate the motion of the particle along its circular arc and determine whether a tumble occurs. To do this, generate a random number R between 0 and 1 and use the condition

$$R < P_{\text{tumble}}$$

to determine whether a tumble occurs. If a tumble occurs, assign the particle a new random orientation θ . Otherwise, the particle continues along its current circular arc.

Store the x -position, y -position, orientation θ , and squared displacement

$$r^2 = x^2 + y^2$$

as functions of time.

Plot a random $x - y$ trajectory of the particle and identify the tumble events on the trajectory. In the trajectory, identify the start and finish and mark the tumbling events.

4.3 Ensemble of Particles

Extend your model from a single particle to an ensemble of N particles. Each particle should undergo its own sequence of random tumble events. Assume these particles do not interact with each other.

Calculate the squared displacement of each particle,

$$r_p^2(t) = x_p^2(t) + y_p^2(t).$$

Use the ensemble to calculate the mean squared displacement,

$$\text{MSD}(t) = \frac{1}{N} \sum_{p=1}^N r_p^2(t).$$

Plot the MSD on a logarithmic scale as a function of time and explain what the result tells you about the spreading of the particles.

4.4 Convergence Study

Here, you will investigate how the numerical solution depends on the simulation parameters. This is essential to verify the independence of a solution from your selection of:

1. the time step Δt ;
2. the number of particles N .

Compare your results and determine values of Δt and N for which the calculated MSD is sufficiently converged.

4.5 Effect of the Tumble Rate

Investigate how the physical behaviour of the system changes with the tumble rate α . Here, you will take the converged solution from the previous task.

Choose at least two additional non-zero values of α and compare:

1. the individual particle trajectories;
2. the MSD as a function of time;
3. use a linear fit for the MSD data. It can be piecewise.

Explain how changing the tumble rate affects the ability of the particles to explore the surrounding space, and how this changes over short and long periods of time.

4.6 Variable Tumble Rate

So far, the tumble rate α has been assumed to be constant. Now, consider a tumble rate that varies with the position of the particle according to

$$\alpha(x) = \alpha_0 + kx,$$

where α_0 is the tumble rate obtained from (Turuban et al., 2025) and k determines how strongly the tumble rate changes with position.

At time step n , the tumble rate is therefore

$$\alpha^n = \alpha(x^n) = \alpha_0 + kx^n.$$

Since the tumble rate depends on the particle position, the probability of a tumble must be recalculated at every time step as

$$P_{\text{tumble}}^n = 1 - e^{-\alpha^n \Delta t}.$$

The random number R is generated in the same way as in the constant tumble-rate model. However, the probability against which R is compared now changes as the particle moves through the domain,

$$R < P_{\text{tumble}}^n.$$

Modify your code to include the spatially varying tumble rate and compare the results with the constant- α case.

Compare:

1. the individual particle trajectories;
2. the MSD as a function of time;
3. use a linear fit for the MSD data. It can be piecewise.

Explain how the spatial variation in the tumble rate affects the ability of the particles to explore the surrounding space, and how this changes over short and long periods of time.

5 Submission Format and Assessment

The project will be assessed through an **oral presentation**. The presentation slides should clearly describe the work completed, the physical and numerical parameter selections and their justification, the simulation results, and the overall architecture and organisation of the code.

The following grading criteria will be used:

- **91–100% — Exceptional:** The project is complete and demonstrates an exceptional level of understanding and implementation. Modelling and numerical choices are clearly explained and well justified. The code is appropriately structured and the results are presented, interpreted, and discussed to a high standard.
- **81–90% — Very Good:** The project demonstrates an above-average level of understanding and implementation. The required tasks are substantially complete and the main modelling and numerical choices are appropriately justified. There may be repeated minor issues, but these do not substantially affect the overall quality of the work.
- **71–80% — Good:** The project demonstrates a good understanding of the main concepts and provides a reasonable implementation of the required tasks. However, there are some substantial issues, omissions, or insufficiently justified choices that affect the overall quality of the work.
- **61–70% — Satisfactory:** The project demonstrates a basic understanding of the required concepts, but contains several gaps, incomplete elements, technical issues, or inadequately justified choices. The implementation and/or interpretation of the results requires substantial improvement.
- **51–60% — Limited:** The project demonstrates limited achievement of the project objectives. Significant parts of the required work are incomplete, incorrect, or insufficiently explained, and there are major gaps in the implementation or interpretation.

Penalties

- **Use of AI:** AI may be used only to assist with syntax. If AI is used to generate the entire code, the maximum grade for the project will be **50%**. Any additional issues identified in the work may reduce the grade further.

- **Late submission:** A late submission will receive an automatic deduction of **10%**.
- **No submission:** Failure to submit the project will result in **no grade**.

References

Turuban, R., Noselli, G., Beran, A., & DeSimone, A. (2025). Cooperative mixing through hydrodynamic interactions in *Stylonychia lemnae*. *Proceedings of the National Academy of Sciences of the United States of America*, 122(37), e2500588122. doi: 10.1073/pnas.2500588122